

## ***Short or Long Sleep Duration Is Associated with Memory Impairment in Older Chinese: the Guangzhou Biobank Cohort Study***

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Study Objectives: To examine the association between sleep-related factors and memory impairment. Design: Cross-sectional study Setting: Community-based study in Guangzhou, China. Participants: 28,670 older Chinese (20,776 women and 7,894 men) aged 50 to 85 years. Measurements and Results: Demographic and socioeconomic data, sleep-related factors, and cognitive function were collected by face-to-face interview. Potential confounders, such as employment and occupational status, smoking, alcohol and tea use, physical activity, self-rated health, anthropometry, blood pressure, and fasting plasma glucose and lipids were measured. After adjusting for multiple potential confounders, an inverted U-shaped association between sleep duration and delayed word recall test (DWRT) score, a validated measure of memory impairment, was found, with 7 to 8 h of habitual sleep duration showing the highest score (P-values for trend from 3 to 7 h and from 7 to  $\geq 10$  h were all  $\leq 0.001$ ). Compared to sleep duration of 7 h, the adjusted odds ratio for memory impairment from the sleep duration of 3 to 4 or  $\geq 10$  h was 1.29 (95% confidence interval 1.07-1.56) and 1.52 (1.25-1.86), respectively. Subjects with daily napping, morning tiredness, or insomnia had significantly lower DWRT scores than those without (P ranged from  $< 0.001$  to 0.01). Conclusions: Short or long sleep duration was an important sleep-related factor independently associated with memory impairment and may be a useful marker for increased risk of cognitive impairment in older people. Citation: Xu L; Jiang CQ; Lam TH; Liu B; Jin YL; Zhu T; Zhang WS; Cheng KK; Thomas GN. Short or long sleep duration is associated with memory impairment in older Chinese: the Guangzhou Biobank Cohort Study. SLEEP 2011;34(5):575-580.